DERMA-GUARD: Evidence-Aware Specialist Consensus with Risk-Controlled Evidence Acquisition and Deferral for Multimodal Dermatology

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Abstract—Skin-lesion classifiers rarely fail for lack of accuracy on clean benchmarks. They fail because they answer every case with the same apparent confidence: a blurred photograph, missing history, or a photograph and a history that point to different diagnoses all produce a label. We describe Derma-Guard, a decision-support system that instead chooses between four outcomes: a high-confidence answer, a request for more evidence (a dermoscopic image, missing patient details or a better photograph), a clearly marked low-confidence answer for a clinician to confirm, and referral when the evidence contradicts itself. Six specialists (evidential heads for the photograph, dermoscopy and metadata, an external legacy model, a nearest-neighbour vote and a class-balanced embedding classifier) are profiled case by case by an evidence manager, and a learned gate decides how much each should count; the gate was chosen from seven combination rules under five-fold cross-fitting. A calibrated, cost-aware policy with a Learn-then-Test bound on missed cancers makes the final choice. On 787 held-out MILK10k lesions, 89.1% of patients receive an answer and 57.1% a high-confidence one that is 77.1% accurate over eleven diagnoses and misses 3.1% of cancers, against 4.9% for a learned rejector ($p = 0.011$) and 3.8% for confidence thresholding at the same coverage. Across four fusion back-ends the policy's miss rate stays between 3.0% and 3.5%, while thresholding ranges from 3.5% to 9.3%: abstention by confidence is only as safe as the probabilities beneath it. Controlled image damage shows that evidential uncertainty misses blur, darkness, noise and occlusion; explicit checks remove every confident answer on damaged input. Without retraining, the system stays within its safety target on PAD-UFES-20 and Derm7pt, while its fine-grained diagnosis does not transfer. Re-evaluating our earlier prototype, whose reported accuracies of 84–90% were not reproducible, places its models at 57–75% on unseen patients.

Index Terms—dermatology, multimodal fusion, evidential deep learning, selective prediction, conformal prediction, risk control, evidence acquisition, clinical decision support

I. Introduction

A patient who photographs a mole and asks an app whether it is dangerous gets an answer almost every time. The answer comes with a number, often above 90%, and that number looks the same whether the photograph was sharp or blurred, whether the patient's age and the lesion's location agree with the image, and whether the lesion resembles anything in the training data. Deep networks reached dermatologist-level accuracy on curated images years ago [1], so accuracy on benchmarks is no longer the main obstacle to safe use. Behaviour is.

A careful dermatologist behaves differently. When the evidence is clear they give a diagnosis. When it is not, they look more closely with a dermatoscope, give a provisional opinion and ask for confirmation, or refer the patient when the history and the lesion tell different stories. We wanted to know whether a model can be built to behave this way, and how much it costs.

This paper is also the second attempt. The first version of Derma-Guard already had the right intuitions (a panel of specialist models, a sceptical component that questions their evidence, and trust that depends on the suspected diagnosis), but its weights were typed in by hand. When we audited it, its reported accuracies did not survive: the split was made by photograph rather than by patient, and its evaluation code adjusted predictions toward preset accuracy targets (Section V-E). We kept the ideas, replaced every hand-set number with one fitted on validation data, and made every result a script that can be re-run.

Our two base papers (Section II) are HOT AI [2], the closest task, which asks for a dermoscopic image when the

photograph is not enough, and Learn-then-Test [3], which provides the statistical guarantee. We compare against both, or against a faithful baseline built from them, on the same probabilities.

Contributions. (i) A six-specialist team whose per-case weights are learned from all of the evidence manager's signals, chosen from seven rules by cross-fitted validation, with an exhaustive subset study and an oracle ceiling. (ii) A calibrated four-outcome policy that can request three kinds of missing evidence and certifies the missed-cancer rate of its confident answers. (iii) The finding that confidence-based abstention is only as safe as the fusion beneath it, whereas the calibrated policy is stable across fusion back-ends. (iv) A controlled degradation benchmark, two external tests without retraining, and an honest re-evaluation of the earlier prototype's seven models. (v) A working care platform, with every system variant available side by side in its interface.

II. Base Papers and Related Work

Base paper 1: HOT AI [2] is the closest task. From a clinical image it produces a hierarchical diagnosis, an out-of-distribution alert and a recommendation to take a dermoscopic image when the photograph alone is not enough, and then fuses both images. It establishes that an AI which asks for dermoscopy is not new. What Derma-Guard adds is an explicit cost for acquiring and for deferring, a separate defer action for contradictory evidence, a certified bound on missed cancers, and stress tests on a public paired dataset. Because HOT AI's model and data are not public, we compare against a HOT-style rule built from it: ask the most uncertain patients for dermoscopy, at the same rate as our policy, on the same probabilities.

Base paper 2: Learn-then-Test [3] chooses decision thresholds on held-out data so that a chosen risk is bounded with a finite-sample guarantee when future cases resemble the calibration cases. We use it unchanged for the risk that matters here, the share of answered cancers called benign, inside a three-action policy, and we measure what happens to the guarantee on two other datasets.

Data and evaluation. MILK10k [4] pairs a clinical close-up and a dermoscopic image for every lesion; a reader study on it [5] found that dermoscopy has higher sensitivity for malignancy and the close-up higher specificity, each adding value, which is the clinical reason to ask for dermoscopy selectively. JI-ADF [6] reports multimodal fusion on MILK10k's hidden test, which we did not use, so its numbers are not comparable with ours. We evaluate abstention with AUGRC [7] and matched-coverage comparisons, and acquisition against random and oracle bounds in the spirit of CAMA [8].

Classification and fusion. Deep networks reached dermatologist level on curated images [1], typically on residual networks [9]; dermatology foundation models [10] are the frontier. Multimodal work fuses images with metadata [11] or pairs clinical and dermoscopic images [12]; PAD-UFES-20 [13] provides smartphone photographs with metadata. These models improve the prediction that a policy sits on; fusion is a component here, not the contribution.

TABLE I
The two base papers (top) and the other methods we compare against, with what we measured (MILK10k test, $c_{\text{miss}} = 5$).

Work	Its contribution	Our use and measured result
HOT AI [2]	asks for dermoscopy when the photo is insufficient; OOD alert	adds costs, defer action, certified bound; HOT-style "ask the most uncertain" rule: acquisition efficiency +0.26 vs +0.07 for our estimator (it wins), but answering everyone afterwards misses 5.6% vs 3.1% for our confident answers
Learn-then-Test [3]	risk-controlling thresholds	certified at both stages ($\hat{\lambda}$ 0.218, 0.276); held on PAD-UFES-20 (0.0%) and Derm7pt (3.8%)
Learning to defer [25]	learned deferral	L2D-style rejector misses 4.9% vs our 3.1% ($p = 0.011$)
Confidence threshold [24]	selective classification	3.8% vs 3.1% (n.s.); 3.5–9.3% across four fusions
AUGRC [7]	abstention metric	ignorance 0.0820 vs softmax 0.0757 (softmax better)
MILK10k [4], [5]	paired data; each modality adds value	dermoscopy head AUROC 0.854 vs photo 0.787; fused 0.885

Uncertainty, conformal sets and deferral. Evidential deep learning [14] and Dempster–Shafer fusion [15], [16] give per-case ignorance and conflict; temperature scaling [17], isotonic regression [18] and Mahalanobis scores [19] calibrate and detect unusual inputs. Conformal prediction has been validated for skin lesions [20], made group-conditional by skin type [21], and is smallest on average in its least-ambiguous form [22]. Chow's rule [23], selective classification [24] and learning to defer [25] decide when not to answer. Each ingredient exists; the open question is whether combining a cost model, a statistical guarantee, a three-action policy and combined-failure testing on public paired data measurably helps.

III. Data and Evaluation Protocol

MILK10k [4] has 5,240 lesions, each with one clinical close-up and one dermoscopic image, eleven diagnoses (95.7% confirmed by histopathology; 71.7% malignant), age, sex, body site, a skin-tone class and seven automatically computed concept scores. Without patient identifiers, we split by lesion into 3,667 training, 786 validation and 787 test lesions, asserting that no lesion appears twice. Validation fits everything that is fitted after the network: fusion rules, reliabilities, calibration, conformal quantiles, the risk threshold, the acquisition estimator and every choice between methods. The test split was scored once. The data are heavily imbalanced: basal cell carcinoma makes up 48% of lesions, while four classes have fewer than ten test lesions each, which is

why we report macro-F1 and balanced accuracy next to accuracy throughout.

External data. PAD-UFES-20 [13] (2,298 smartphone photographs, six classes, no dermoscopy) and Derm7pt [12] (1,003 lesions with both images after mapping its diagnoses onto MILK10k classes) are used only for testing, without retraining or recalibration.

Metrics. For a system that may abstain we report coverage (share of high-confidence answers), accuracy among them and the missed-cancer rate (share of answered malignant lesions called benign), always comparing systems at equal coverage; macro-F1 and balanced accuracy for the rare classes; malignant-vs-benign AUROC; AUGRC [7]; acquisition efficiency $\mathrm{AE} = (G_\pi - G_{\mathrm{rand}})/(G_{\mathrm{oracle}} - G_{\mathrm{rand}})$ for the choice of whom to send for dermoscopy; and 95% bootstrap intervals with paired tests (2,000 resamples).

IV. Method

Fig. 1 shows the pipeline. A case moves left to right: specialists give opinions, the evidence manager profiles them, a fusion rule turns the profile into weights, and the policy decides.

A. Specialists

S1–S3: evidential heads. An ImageNet ResNet50 [9] shared by both image types and a residual metadata network produce three 256-dimensional embeddings, each feeding an evidential head with non-negative evidence e_m [14]. With $\alpha_m = e_m + 1$ and $S_m = \sum_k \alpha_{m,k}$, belief and ignorance are $b_{m,k} = e_{m,k}/S_m$ and $u_m = K/S_m$; a head contributes its pignistic distribution $p_m = b_m + u_m/K$. The heads are also combined by Dempster's rule [16],

$$C = \sum_{i \neq j} b_{1,i} b_{2,j}, \quad b_k = \frac{b_{1,k} b_{2,k} + b_{1,k} u_2 + b_{2,k} u_1}{1 - C}, \quad u = \frac{u_1 u_2}{1 - C}, \tag{1}$$

which leaves a missing modality ($u = 1$) exactly without effect and separates contradiction (C) from shared ignorance (u). The network was trained on a Kaggle GPU for 25 epochs with modality dropout 0.25, so that it learns to say "I don't know" for an absent input.

S4: legacy model. The earlier prototype's image model, trained on PAD-UFES-20, is applied unchanged; its six classes are mapped to MILK10k and mixed with 10% uniform mass so unsupported classes are unknown rather than impossible.

S5, S6: case-based specialists. S5 is a vote of the 15 most similar training lesions in embedding space, weighted by $\exp(5\,\mathrm{sim})$. S6 is a multinomial logistic regression over all training embeddings with class weights inversely proportional to frequency, so that rare diagnoses are not drowned out; its regularisation was chosen on validation.

B. The evidence manager

For specialist m on case x the evidence manager computes ten signals: quality Q_m (share of image statistics inside the training 5th–95th percentiles), a reliability R_m (logistic regression fitted on validation predicting whether m is right), normalised entropy U_m, confidence, margin, agreement with the other specialists, the measured track record $\pi_m(\hat{y}_0) = \Pr(m \text{ correct} \mid y = \hat{y}_0)$ at the provisional diagnosis, the team's Jensen–Shannon consistency, missingness, and the Mahalanobis out-of-distribution distance [19]. The earlier prototype multiplied hand-set versions, $E_m = Q_m R_m U_m C S M_m$, and normalised them into weights; because consistency C and sufficiency S are equal for all specialists of a case, they cancelled, and only three signals ever moved the weights.

C. Fusion rules and how one is chosen

Iterative diagnosis-adaptive fusion (v1) keeps the prototype's EM-style loop [26]: weights $w_m \propto (Q_m R_m)^a$ give a provisional diagnosis $\hat{y}$; then up to three times $w_m = \mathrm{softmax}_m\big(a \log Q_m R_m + b \log \pi_m(\hat{y}) + g \log \kappa_m\big)$, with $\kappa_m = \rho$ for specialists disagreeing with $\hat{y}$, and the diagnosis is updated.

The learned gate (v2) lets every signal act. With the standardised signal vector $\phi_m(x)$,

$$w_m(x) = \underset{m \in \mathrm{present}}{\mathrm{softmax}} \big(\beta_m + \theta_m^\top \phi_m(x)\big), \tag{2}$$

and the opinions are pooled either as a mixture $\sum_m w_m p_m$ or as a product of experts $\propto \prod_m p_m^{w_m}$, followed by temperature scaling. θ, β and the temperature minimise validation log-loss with an ℓ_2 penalty.

Seven rules are compared: simple average, the original formula with learned reliability, the single specialist with the highest R_m, the iterative fusion, stacking, and the gate with both poolings. Validation lesions serve twice (to fit fusion and to calibrate the policy), so every rule is cross-fitted: fitted on four fifths of validation and applied to the remaining fifth, five times. The out-of-fold probabilities are used both to select the rule (best macro-F1, fixed before any test result was seen) and to calibrate the policy. We also refit the selected rule on all 31 (stage A) and 63 (stage B) specialist subsets, and report an oracle that picks, per case, the specialist giving the true class most mass. Fig. 3 summarises the protocol.

D. The decision policy

Stage A uses the photograph and metadata; stage B adds dermoscopy. The stated diagnosis is the most probable class; costs are 0 for a correct answer, 1 within the benign or malignant group, $c_{\mathrm{miss}} = 5$ for a cancer called benign and 2 for the reverse, with 0.5 to acquire dermoscopy and 1.5 for a clinician review. Gates are: Dempster–Shafer conflict above its validation 90th percentile (Defer); a Mondrian least-ambiguous conformal set over {benign, malignant} ($\alpha = 0.1$) [21], [22] that still contains "malignant" when the answer would be benign; a Learn-then-Test threshold [3] on $1 - u$ such that $\Pr(\text{missed-cancer rate} \leq 0.05) \geq 0.9$; severe image-quality issues (blur, exposure, contrast, glare, grain, or a covered image centre beyond the training 1st/99th percentiles); a Mahalanobis outlier; and missing details (no age, sex or

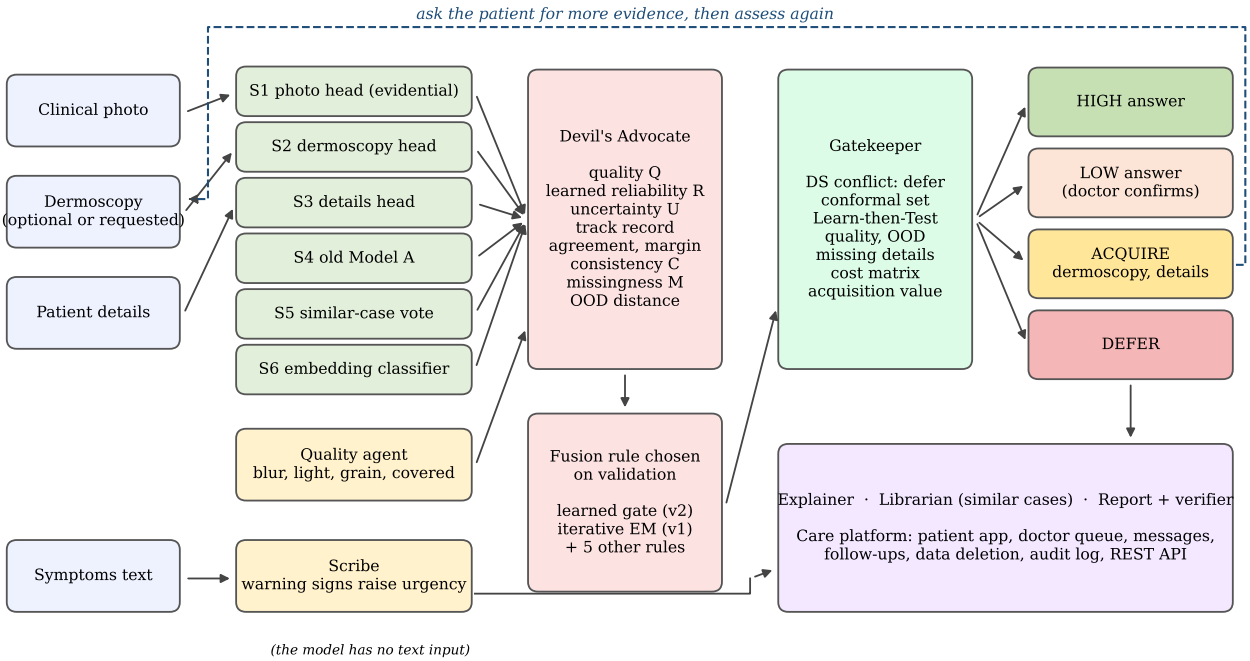


Fig. 1. Derma-Guard. Six specialists are profiled by the evidence manager ("Devil's Advocate") and combined by the fusion rule chosen on validation; the gatekeeper and a cost matrix choose one of four outcomes. The dashed loop is the request for more evidence, after which the case is assessed again.

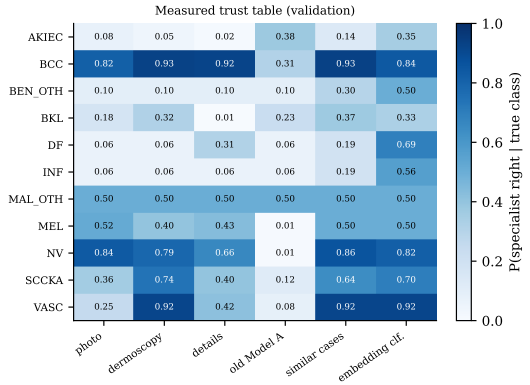


Fig. 2. The measured trust table $\pi_m(c)$: on validation lesions of each true class, how often each specialist alone names it. The prototype typed this table in by hand; for melanoma it assumed 0.96 for the photograph, where 0.52 is measured.

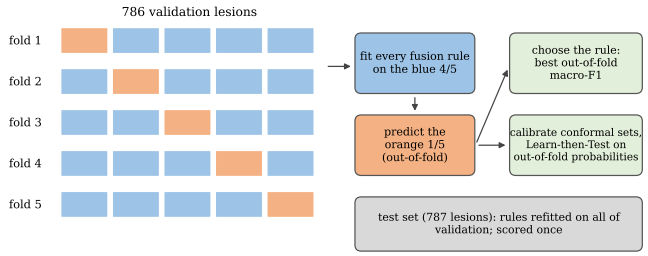


Fig. 3. Cross-fitting. Every fusion rule is fitted on four validation folds and predicts the fifth; the out-of-fold predictions both choose the rule and calibrate the safety guarantees, so neither step judges a model on its own training data.

site and no dermoscopy), which triggers a request instead of an answer. If no gate fires and answering is cheapest, the answer is High. Otherwise the system requests dermoscopy when the expected cost after acquisition, estimated by a gradient-boosted regressor fitted on validation lesions whose dermoscopy is known, plus 0.5, is below the cost of review; else it gives a Low answer for a clinician to confirm. Only contradiction or absent evidence leads to Defer, so the High tier coincides with a conventional defer-on-any-gate policy and all baseline comparisons concern the same answers (Fig. 4). An earlier version deferred on every failed check, which sent every patient without a confident answer to a clinician with nothing at all; the Low tier keeps those patients informed while still requiring confirmation.

E. Agents and platform

Around the policy, a quality agent advises on retakes, a scribe raises triage urgency when the patient's own words mention warning signs (with negation handling; it never changes the diagnosis), a librarian shows similar confirmed training lesions , an explainer states who said what, why dissenting specialists disagree and where the weight went,

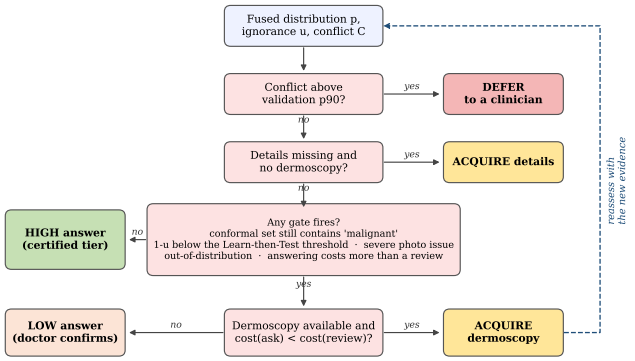


Fig. 4. How the policy chooses an outcome. Requests for evidence loop back: once the dermoscopic image or the missing details arrive, the case is assessed again.

and a report writer may use a local language model whose output is discarded if a verifier finds any diagnosis or percentage not in the evidence. A care platform built on SQLite routes each outcome: a confident answer schedules a follow-up, a request moves the case to "awaiting patient" until the evidence arrives, and Low answers and deferrals enter a clinician queue ordered by kind and urgency. A follow-up photograph whose probability of malignancy rises by 0.15 or more returns the case as urgent. Every assessment stores the policy version that produced it, every event goes to an append-only audit log, and patients can delete a case or their whole record, including uploaded photographs. A Streamlit application exposes patient, clinician, stress-testing and research views, and a "compare systems" view runs any held-out lesion through every specialist, every fusion rule and every baseline side by side; a FastAPI service exposes the same workflow. The code base has 230 automated tests.

For one held-out melanoma in our test set, for instance, the explainer reports that the photograph head named melanoma but was only 10% sure, that the legacy model said benign keratosis because it "was trained on a different dataset and cannot name five of the eleven diagnoses" and was right on only 1% of validation melanomas, and that most of the weight went to the metadata head, the embedding classifier and the photograph head, which agreed with the team and had high learned reliability for this case. The system then asked for the dermoscopic image. Every number in such an explanation is read from the evidence profile; the language model, when used, may only reword it.

V. Results

A. Specialists and fusion

Table II and Fig. 5 summarise the specialists and the fusion rules. Dermoscopy carries real information: the dermoscopy head beats the photograph head on every metric, and adding it lowers the network's mean ignorance

TABLE II
Specialists alone and the main fusion rules on the test split (stage B: with dermoscopy). Val: out-of-fold validation macro-F1 used for selection.

	Val	Acc.	F1	Bal.	AUROC
Specialists (stage B)					
S1 photograph head	–	0.595	0.264	0.266	0.787
S2 dermoscopy head	–	0.705	0.400	0.411	0.854
S3 metadata head	–	0.600	0.246	0.252	0.749
S4 legacy model	–	0.219	0.077	0.096	0.553
S5 nearest-neighbour vote	–	0.719	0.431	0.402	0.866
S6 embedding classifier	–	0.700	0.440	0.447	0.875
Fusion, stage A (photograph + metadata)					
Dempster–Shafer, heads	0.274	0.644	0.286	0.280	0.799
Simple average	0.344	0.637	0.343	0.338	0.824
Iterative EM (v1)	0.374	0.670	0.383	0.375	0.832
Stacking	0.272	0.644	0.308	0.296	0.822
Learned gate (v2)*	0.412	0.657	0.377	0.360	0.838
Fusion, stage B (+ dermoscopy)					
Dempster–Shafer, heads	0.361	0.690	0.376	0.378	0.856
Simple average	0.457	0.714	0.422	0.400	0.870
Iterative EM (v1)	0.492	0.717	0.459	0.444	0.862
Stacking	0.413	0.720	0.372	0.361	0.853
Learned gate (v2)*	0.517	0.731	0.438	0.418	0.885
Oracle best specialist	–	0.840	0.594	–	–

*selected (product pooling). All seven rules: Fig. 5.

from 0.676 to 0.439. The legacy model transfers poorly (21.9% accuracy). The two case-based specialists are the strongest single sources, and the class-balanced classifier S6 has the best balanced accuracy.

Every way of weighting the team that uses the evidence beats the evidential network alone: without dermoscopy, macro-F1 rises from 0.286 to 0.377 for the selected gate, and with dermoscopy the gate has the best accuracy (0.731) and AUROC (0.885). The gate won the cross-fitted validation comparison in both stages; on test the iterative fusion has a slightly higher macro-F1, a reminder that macro-F1 over classes with fewer than ten test lesions is noisy. We kept the pre-declared choice. Stacking, the textbook learned combiner, is the weakest learned rule: 66 free inputs and 786 validation lesions are not enough. Adding S6 raised the selected rule's stage-A macro-F1 from 0.332 to 0.377, and the subset study found no subset that beat the full team by more than 0.033 on validation. The oracle's 0.594 shows that per-case selection, not the specialists, is the bottleneck. On average the gate gives the largest weights to the metadata head (0.35–0.41) and S6 (0.21–0.22), moves weight to dermoscopy when it is present, and gives the legacy model 0.07. Isotonic calibration makes the displayed confidence honest: 0.757 shown on average against 0.731 actual accuracy with dermoscopy, 0.660 against 0.657 without (Fig. 7).

Per class (Fig. 6), the team helps most where the evidential network was weakest: benign keratoses (F1 0.16 to 0.51), actinic keratoses (0.24 to 0.47) and the small "benign other" and inflammatory classes, which the

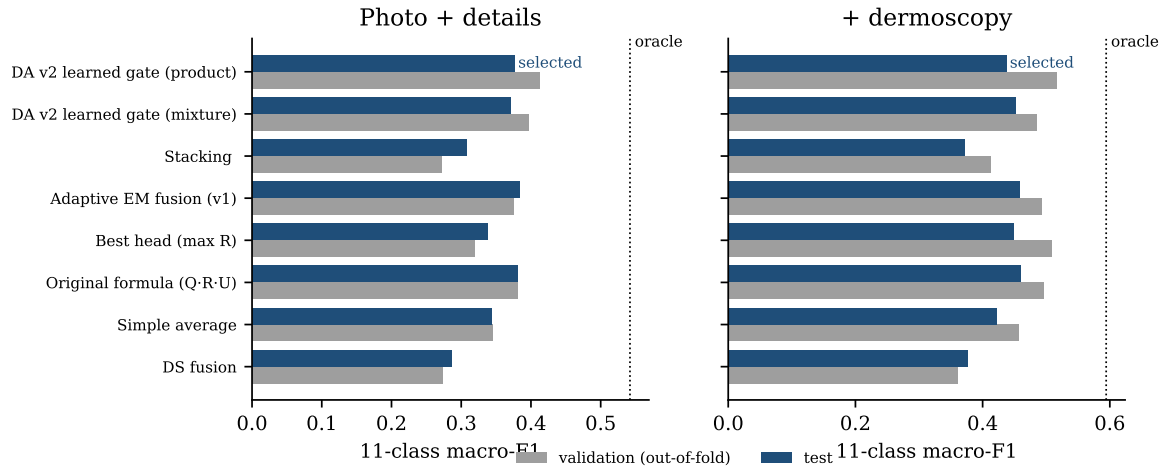


Fig. 5. All seven combination rules: out-of-fold validation macro-F1 (grey, used to select) and test macro-F1 (blue). The dotted line is the oracle that always picks the best specialist.

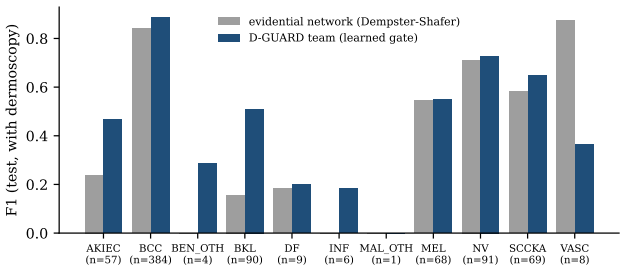


Fig. 6. Per-class F1 on the test set with dermoscopy: the evidential network alone and the full team. n is the number of test lesions.

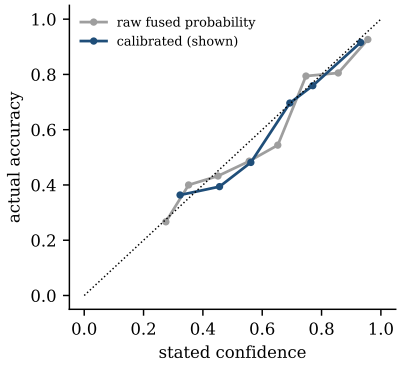


Fig. 7. Reliability diagram on the test set: the calibrated confidence that the application displays tracks the observed accuracy.

network never predicted. It is worse on vascular lesions (0.88 to 0.36 on eight test lesions), where the dermoscopy head alone was the best judge, and the single malignant-other lesion is never found.

B. Decisions

Derma-Guard answers 89.1% of patients (Table III). Its 57.1% high-confidence answers are 77.1% accurate (95% CI 73.0–80.8%) and miss 3.1% of the cancers they answer (1.6–4.8%); Learn-then-Test certified thresholds at both stages ($\hat{\lambda}$ = 0.218, 0.276). Low-confidence answers are right only 49.6% of the time, which is why they always go to a clinician, but the truth is in their top three for 85.7%. Dermoscopy was requested for 29.5% of patients.

TABLE III
Decision policy and baselines on the same probabilities (test, $c_{\text{miss}} = 5$). Abstaining baselines are matched to Derma-Guard's coverage.

Policy	Ans.	Derm.	Acc.	Missed	Cost
Always answer, photograph	100%	0%	0.657	7.2%	0.759
Always acquire, answer	100%	100%	0.731	6.4%	1.124
Confidence threshold (B3)	57.1%	0%	0.815	3.8%	0.881
Learned rejector (L2D)	57.1%	0%	0.764	4.9%	–
Entropy-chosen acquisition	100%	29.5%	0.700	5.6%	0.790
Derma-Guard, High tier	57.1%	29.5%	0.771	3.1%	1.047
Derma-Guard outcomes: High 57.1%, Low 32.0% (top-3 85.7%), Defer 10.9%					

At the same coverage, a learned rejector trained to predict the model's own errors [25] misses 4.9% of cancers, significantly more than Derma-Guard (+1.8 points, CI +0.2 to +3.5, one-sided p = 0.011). Plain confidence thresholding misses 3.8%; the difference (+0.7 points, CI −0.7 to +2.0, $p = 0.18$) is not significant, and thresholding is more accurate (81.5% vs. 77.1%) because it answers only the photograph's easiest cases. Always answering is cheapest in mean cost, since every clinician review is charged 1.5. Fig. 8 places these points on the full risk–coverage curves: no coverage level of the threshold reaches the policy's miss rate at 57% coverage.

Is the gap to thresholding real? We ran the identical policy on four fusion back-ends (Fig. 9). Its miss rate barely moved, 3.0–3.5%. The threshold's moved almost threefold, from 3.5% (the five-specialist gate) to 9.3% (the five-specialist iterative fusion). A confidence threshold is exactly as safe as the ranking it receives, and nobody

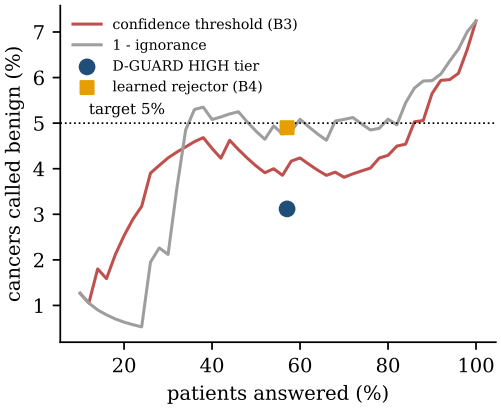


Fig. 8. Share of answered cancers called benign against the share of patients answered, for abstention by confidence and by $1-u$; the points are Derma-Guard's High tier and the learned rejector at the same coverage.

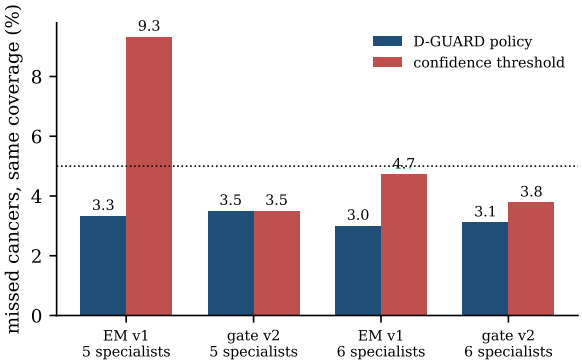


Fig. 9. The identical policy over four fusion back-ends (iterative EM or learned gate, with five or six specialists). The policy's missed-cancer rate stays at 3.0–3.5%; confidence thresholding at the same coverage ranges from 3.5% to 9.3%.

deploying it knows in advance which kind of ranking they have; the calibrated policy delivers the same certified rate on all four. The same comparison contains a result against us: the learned acquisition estimator chose whom to send for dermoscopy far better than entropy on the iterative fusion (AE +0.30 vs. −0.04) but worse on the final gate (+0.07 vs. +0.26), and ignorance ranked errors better than softmax on the former (AUGRC 0.0874 vs. 0.0918) but not the latter (0.0820 vs. 0.0757). Once the fused probabilities are good, simple uncertainty is a good enough guide.

Raising c_{miss} from 2 to 10 lowers the High share from 61.6% to 53.4% and the miss rate from 4.1% to 1.9% (thresholding: 3.8% at the same coverage), while about 89% of patients still get an answer (Fig. 10). The cost of a missed cancer is therefore a dial a deploying clinic can set, not a hidden constant.

Fairness is a weakness (Fig. 11): for the 67 test lesions of skin tone 2 the High miss rate is 11.5%, against 2.2% for tone 3 ($n = 489$); tone 1 has only 14 lesions. MILK10k is dominated by tone 3, and a calibration fitted on the whole

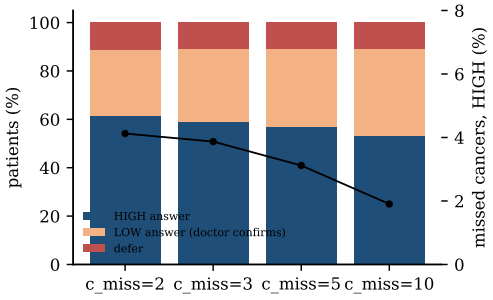


Fig. 10. Outcome shares (bars) and the missed-cancer rate of High answers (line) as the cost of a missed cancer increases.

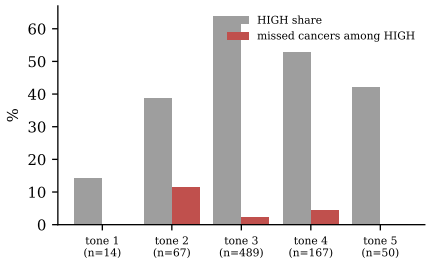


Fig. 11. High share and missed cancers among High answers by MILK10k skin-tone class.

validation set guarantees an average, not every group; group-wise calibration needs more lighter-tone data than the dataset contains.

C. Robustness: DERMA-STRESS

We damaged the photograph of 300 test lesions six ways without retraining (Fig. 12). Evidential ignorance does not catch what matters. Gaussian blur lowers mean ignorance (0.679 to 0.605) and raises the high-confidence rate (46% to 50%); under-exposed and occluded photographs keep 22% and 16% high-confidence answers that miss 12.3% and 14.3% of cancers. Explicit checks do catch it: the quality agent flags every blurred, dark, noisy and occluded photograph (the last two through the grain and covered-centre checks), and every case without metadata becomes a request for it. With these checks no damaged input received a high-confidence answer; JPEG artefacts kept 22% at 1.5% missed. The price is that 8% of clean photographs are also flagged.

What each component adds. Reading the experiments as an ablation: weighting the specialists by their evidence instead of fusing only the evidential heads adds 0.091 photograph-only macro-F1 (0.286 to 0.377); the sixth specialist alone accounts for 0.045 of it; cross-fitting removed an optimistic three points of High coverage that in-sample fitting had produced; the calibrated policy, rather than any particular fusion, is what keeps the miss rate at 3.0–3.5%; and the explicit image and missing-details checks, not evidential ignorance, are what remove confident answers on damaged input.

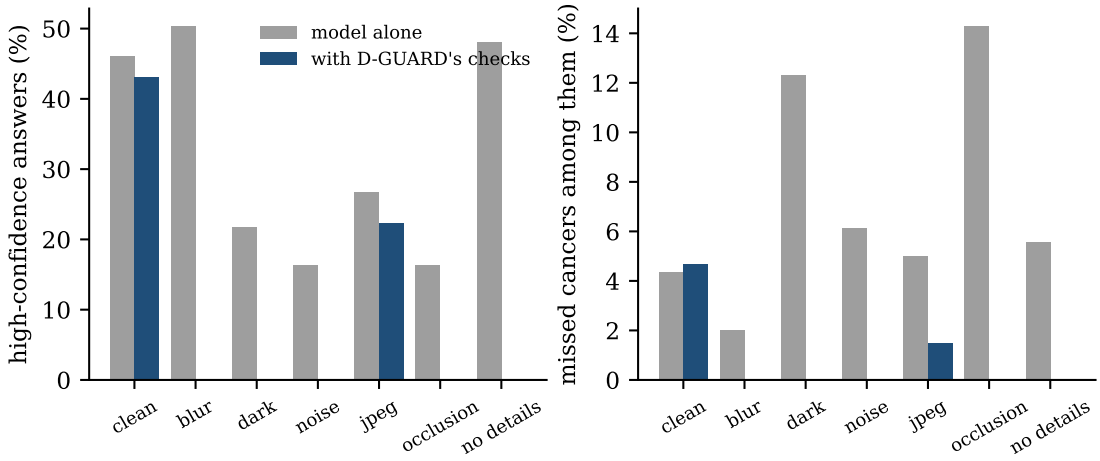


Fig. 12. DERMA-STRESS on 300 test lesions (photograph + metadata): share of high-confidence answers and missed cancers among them, for the model alone (grey) and with the pipeline's quality and missing-details checks (blue).

TABLE IV
The MILK10k system on two other datasets, unchanged.
Threshold: confidence thresholding at the same coverage.

	MILK10k	PAD-UFES-20	Derm7pt
Dermoscopy available	yes	no	on request
Malignant share	72%	79%	29%
Photos failing quality check	8%	28%	29%
Dermoscopy requested	30%	28%	39%
Accuracy, all cases	0.731	0.367	0.479
Malignant AUROC	0.885	0.840	0.738
High share	57.1%	11.4%	13.0%
Accuracy of High answers	0.771	0.517	0.708
Missed cancers, High	3.1%	0.0%	3.8%
Threshold: missed / accuracy	3.8% / 0.815	0.0% / 0.574	5.4% / 0.846

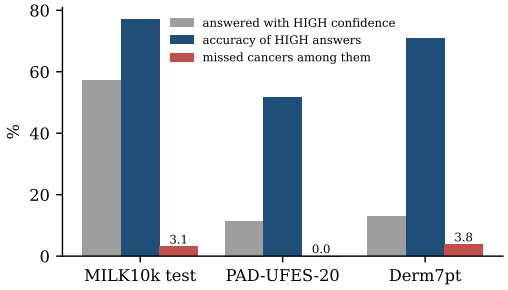


Fig. 13. Share of High answers, their accuracy and the cancers missed among them, on MILK10k and the two external datasets.

D. External tests

The calibration guarantees assume that future patients resemble calibration patients. Table IV shows what happens when they do not. On both external datasets ignorance, quality flags and deferrals rise and the high-confidence share falls from 57% to 11–13%. Among the confident answers that remain the safety target holds: no cancer was missed on PAD-UFES-20 and 3.8% on Derm7pt, below thresholding's 5.4%. On Derm7pt, where dermoscopy exists, the system asked for it for 39% of lesions and overall accuracy rose from 0.402 with photographs alone to 0.479. The eleven-class diagnosis does not transfer: half of PAD-UFES-20's confident answers are wrong, melanoma recall on Derm7pt is 31%, and thresholding is more accurate on both datasets (Fig. 13). Nothing was retrained on either dataset; they are external tests, deliberately kept out of training so that they measure what a deployment would meet.

E. The earlier prototype, re-evaluated

The prototype trained seven PAD-UFES-20 models (image, metadata, free text and their combinations) and

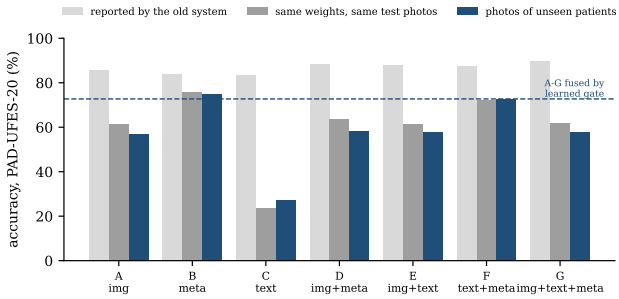


Fig. 14. The earlier prototype's seven PAD-UFES-20 models: accuracy as reported, re-scored with the same weights on the same test photographs, and on photographs of patients never seen in training. Dashed: the seven fused by Derma-Guard's learned gate (cross-validated).

reported 84–90% accuracy. Its split was random by photograph, so 43% of test photographs came from a lesion and 62% from a patient already in training, and its evaluation function swapped probabilities between the true class and its closest rival until the number of correct predictions met a hard-coded target per model. Loading the saved weights and cached features unchanged and scoring them

TABLE V
The prototype's models on PAD-UFES-20: reported accuracy and honest re-scoring of the same saved weights.

	Inputs	Reported	Same test	Unseen patients
A	image	85.5%	61.4%	56.8%
B	metadata	84.1%	75.9%	75.0%
C	free text	83.5%	23.5%	27.3%
D	image + metadata	88.4%	63.5%	58.3%
E	image + text	88.1%	61.4%	57.6%
F	text + metadata	87.2%	72.2%	72.7%
G	all three	89.9%	62.0%	57.6%
A–G fused by the learned gate (unseen patients)				72.7%
A–G simple average (unseen patients)				59.8%

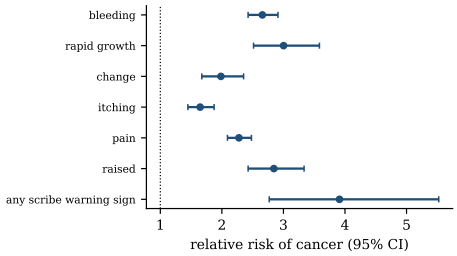


Fig. 15. Relative risk of cancer with each warning sign on the PAD-UFES-20 questionnaire (95% CI). All exceed one.

honestly (Fig. 14): on the 132 test photographs of unseen patients, the image model reaches 56.8%, the metadata model 75.0%, the text model (whose "narratives" were generated from the metadata) 27.3%, and the three-modality model 57.6%, worse than metadata alone. Fusing the seven with our learned gate, cross-validated on those 132 photographs, gives 72.7% against 59.8% for averaging them; the gate recovers almost all of the best model's accuracy without being told which model that is.

Finally, the PAD-UFES-20 questionnaire lets us test the scribe's triage rule (Fig. 15): of 1,641 lesions, those with any of its warning signs were malignant (BCC, SCC or melanoma) in 48.7% of cases against 12.4% without (relative risk 3.9, 95% CI 2.8–5.5), and every single sign had a relative risk between 1.65 and 3.0. A raised lesion, which the scribe did not originally look for, carried a relative risk of 2.8 and was added as a result. These are clinician-collected questionnaire answers rather than free text, so the analysis supports the rule's direction, not its exact wording.

VI. Discussion

What the earlier design got right. Consulting several specialists, questioning their evidence, and trusting them according to the suspected diagnosis all survived honest evaluation; every evidence-aware rule beat both the evidential network alone and plain averaging, and the measured trust table confirms clinical intuition (dermoscopy is reliable for squamous lesions, the photograph for melanoma). What it got wrong was its evidence.

What did not work. Stating the minimum-cost diagnosis (classic Chow [23]) labelled 98.5% of lesions malignant on this 72%-malignant population; adaptive prediction sets for the safety gate collapsed to the full set and deferred every patient; post-hoc logit adjustment raised validation macro-F1 from 0.36 to 0.44 but lowered test macro-F1 from 0.376 to 0.359; fitting the fusion in-sample on the calibration lesions inflated the high-confidence share by three points before cross-fitting; and classes with fewer than ten test lesions are almost never predicted.

Where the gain comes from. The oracle that picks the best specialist per case reaches 0.594 macro-F1, against 0.438 for the learned gate. The specialists already contain much of the right answer; deciding whom to believe is the hard part. A gate trained on more validation data, or distilled from the oracle, is the most direct way forward, and a dermatology foundation backbone [10] would lift every specialist.

Lessons about evaluation. Split by the unit you will meet in practice; never post-process predictions with the label; choose on validation and keep the choice when the test disagrees; and compare against simple baselines on the same probabilities, and on more than one model. The most useful finding of this work, that thresholding's safety is borrowed from the fusion beneath it, appears only under that last rule.

Limitations. All results come from one training seed of an ImageNet ResNet50; 786 validation lesions do a great deal of fitting; MILK10k is a referral population without patient or centre identifiers; the costs are declared, not derived; the guarantees assume exchangeability; the image checks are heuristics; lighter skin tones are under-served; and the system has not been evaluated in a clinic.

VII. Conclusion

Derma-Guard turns a team of imperfect specialists into calibrated clinical actions: it answers nine in ten patients, more than half with a confident answer that misses about one cancer in thirty, asks for dermoscopy, details or a better photograph when they are needed, and refers only when the evidence contradicts itself. Learning how much to trust each specialist from all of its evidence improves photograph-only diagnosis by a third. The calibrated policy is safer than a learned rejector and, unlike confidence thresholding, stays safe whichever fusion it is given. We hope the most transferable lesson is methodological: report the simple baseline on the same probabilities, on more than one model, and keep every number reproducible.

Reproducibility

Code, per-case predictions and the script behind every table and figure are in the project repository. Model weights and images are not redistributed; MILK10k is available from the ISIC Archive under CC-BY-NC.

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